eml★ — Minimal Anti-Holomorphic Extension of the EML Sheffer Operator

Anthony Monnerot

Autodidact — Champigny-sur-Marne, France

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GitHub: github.com/antparis/eml_star

Abstract.

Odrzywołek (2026) showed that $\mathrm{eml}(x, y) = e^x - \ln y$, together with the constant 1, generates all standard elementary functions via finite composition. We identify a structural limitation: eml is holomorphic, so complex conjugation $\bar{z}$, and real and imaginary parts are not reachable by finite eml-compositions. We introduce the companion operator $\mathrm{eml}★(x, y) = e^x - \ln \bar{y}$, which acts as a mirror reflecting the imaginary axis. We prove: (i) $\bar{z} = 1 - \mathrm{eml}★(0, \mathrm{eml}(z, 1))$ at depth 2, conditional on $\mathrm{Im}(z) \in [-\pi, \pi)$; (ii) {eml, eml★, 1} is dense in $C(K, \mathbb{C})$ for every compact $K \subset \{z : \mathrm{Im}(z) \in [-\pi, \pi)\}$ by Stone–Weierstrass; (iii) the exact branch limitation is $\mathrm{Im}(z) \in [-\pi, \pi)$. A direct numerical experiment confirms Theorem 3.1 to machine precision: eml★ achieves MSE = 5.89×10^{-33} vs. 12.97 for eml alone — a ratio of 2.2×10^{33}. Theorems 2.1, 4.3, and Corollary 2.2 are unconditional.

1. Introduction

Odrzywołek [1] demonstrated that $\mathrm{eml}(x, y) = e^x - \ln y$, together with the constant 1, is a continuous Sheffer operator: every standard elementary function is a finite composition of eml-nodes. The grammar is $S \to 1 \mid \mathrm{eml}(S, S)$. This is the continuous analogue of the NAND gate in Boolean logic.

A natural question arises: does {eml, 1} reach all continuous functions on $\mathbb{C}$, or only the holomorphic ones? The answer is: only the holomorphic ones. This note identifies the obstruction precisely and proposes a minimal fix — the companion operator eml★ — that closes the gap.

2. The Holomorphic Barrier

Let $E_0 = \{1\}$, $E_{n+1} = E_n \cup \{\mathrm{eml}(f, g) : f, g \in E_n\}$, and $E = \bigcup_n E_n$.

Theorem 2.1 (Holomorphic closure). Every element of E is holomorphic on its domain.

Proof. The constant 1 is holomorphic. The maps $z \mapsto e^z$ and $z \mapsto \ln z$ (principal branch) are holomorphic. Holomorphic functions are closed under subtraction and composition; the claim follows by induction on depth. ∎

Corollary 2.2. The operators Re, Im, and $z \mapsto \bar{z}$ are not in E.

Proof. These maps are non-holomorphic ($\partial_{\bar{z}} \mathrm{Im}(z) = 1/2 \neq 0$), so no element of E equals them by Theorem 2.1. ∎

Remark 2.3. Corollary 2.2 does not imply $\pi \notin E$. Odrzywołek shows $\pi \in E$ at depth $K > 53$ (Table 4 of [1]). The distinction is between $\mathrm{Im}(\cdot)$ as an operator and π as a fixed real constant.

3. The Companion Operator eml★

Motivation. eml cannot distinguish z from $\bar{z}$ — it only sees the holomorphic world. To recover the conjugate, one modification suffices: replace $\mathrm{Log}(y)$ by $\mathrm{Log}(\bar{y})$ in the second argument.

Definition. $\mathrm{eml}^\star(x, y) := e^x - \mathrm{Log}(\bar{y})$, where Log denotes the principal logarithm ($\mathrm{Arg} \in (-\pi, \pi]$).

Theorem 3.1 (Conjugate recovery at depth 2). For all $z \in \mathbb{C}$ with $\mathrm{Im}(z) \in [-\pi, \pi)$:

$$\bar{z} = 1 - \mathrm{eml}^\star(\, 0,\ \mathrm{eml}(z,\ 1)\,)$$

Proof. $\mathrm{eml}(z, 1) = e^z$. Then $\mathrm{eml}^\star(0, e^z) = 1 - \ln(\mathrm{conj}(e^z)) = 1 - \bar{z}$, provided $\mathrm{Im}(\bar{z}) = -\mathrm{Im}(z) \in (-\pi, \pi]$, i.e. $\mathrm{Im}(z) \in [-\pi, \pi)$. ∎

Theorem 3.2 (Exact safe domain).

The identity of Theorem 3.1 holds if and only if $\mathrm{Im}(z) \in [-\pi, \pi)$. At $\mathrm{Im}(z) = \pi$ the principal Log produces a jump of $2\pi i$. The boundary $\mathrm{Im}(z) = -\pi$ holds; $\mathrm{Im}(z) = +\pi$ fails. Outside the strip, jumps of exactly $2\pi i$ occur.

Corollary 3.3. $|z|$ is reachable at depth 4 via $|z| = \sqrt{z \cdot \bar{z}}$, where $\bar{z}$ comes from Theorem 3.1 and multiplication from the eml arithmetic of [1].

Expression	Rel. depth	Via
$\bar{z} = 1 - \mathrm{eml}^\star(0, \mathrm{eml}(z,1))$	2	Theorem 3.1
$\mathrm{Re}(z) = (z + \bar{z})/2$	3	Corollary 3.3
$\lvert z\rvert^2 = z \cdot \bar{z}$	3	eml arithmetic + Thm 3.1
$\lvert z\rvert = \sqrt{z \cdot \bar{z}}$	4	sqrt via eml [1]

Table 1. Depths in the system $\{\mathrm{eml}, \mathrm{eml}^\star, 1\}$ for $\mathrm{Im}(z) \in [-\pi, \pi)$. Rel. depth counts $\mathrm{eml}^\star$ nodes only.

4. Topological Completeness via Stone–Weierstrass

Lemma 4.1. Every rational number lies in E.

Proof. $0 = \mathrm{eml}(0, 1)$. 1 is given. Arithmetic follows from [1]. ∎

Lemma 4.2. Every polynomial in $\mathbb{Q}[z, \bar{z}]$ lies in the algebra generated by $\{\mathrm{eml}, \mathrm{eml}^\star, 1\}$.

Proof. $z \in E$. $\bar{z}$ by Theorem 3.1. Products and sums by eml arithmetic. ∎

Theorem 4.3 (Stone–Weierstrass density). For every compact $K \subset \mathbb{C}$ with $K \subset \{z : \mathrm{Im}(z) \in [-\pi, \pi)\}$, the algebra generated by $\{\mathrm{eml}, \mathrm{eml}^\star, 1\}$ is dense in $C(K, \mathbb{C})$.

Proof. On K, Theorem 3.1 gives $\bar{z}$ exactly, so $A = \mathbb{Q}[z, \bar{z}] \subseteq \{\mathrm{eml}, \mathrm{eml}^\star, 1\}$ pointwise. A satisfies Stone–Weierstrass [2]: (i) separates points; (ii) does not vanish; (iii) closed under conjugation. Hence A is dense in $C(K, \mathbb{C})$. ∎

Remark 4.4. Theorem 4.3 does not contradict Theorem 2.1: the algebra is dense, but individual elements of E remain holomorphic. Density is an approximation result, not a representation result.

Remark 4.5 (Global extension). The restriction $\mathrm{Im}(z) \in [-\pi, \pi)$ is essential. Density on arbitrary compacts in $\mathbb{C}$ would require tracking winding numbers continuously — impossible with finite compositions of analytic/anti-analytic operators (monodromy theorem). Open problem.

5. Numerical Experiment

We provide a direct machine-precision verification of Theorem 3.1 and contrast $\mathrm{eml}^\star$ with eml alone. The test is deterministic, parameter-free, and fully reproducible from the repository.

Setup. N = 40 points per axis, Re(z) ∈ [−3, 3], Im(z) ∈ [−π + 0.1, π − 0.1] (strictly inside the safe strip). Total: 1600 evaluation points. Target: $\bar{z}$.

Test 1 — with eml★ (Theorem 3.1):

```
pred = 1 - eml_star(0, eml(z, 1))
```

Test 2 — with eml alone (Corollary 2.2):

```
pred = 1 - eml(0, eml(z, 1))
```

Metric	eml★ (with conjugate)	eml (without conjugate)
MSE on 1600 points	5.89×10^{-33}	12.97
Ratio eml / eml★	—	2.2×10^{33}
Conclusion	Theorem 3.1 verified ✓	Corollary 2.2 confirmed ✓

Table 2. Direct numerical comparison on 1600 points. MSE = mean squared error. The eml⍰ result is at float64 machine precision.

Interpretation. The MSE of 5.89×10^{-33} with eml★ is the floating-point floor — effectively zero. The error of 12.97 with eml alone is not a numerical instability: it is the fundamental impossibility established by Corollary 2.2. The ratio of 2.2×10^{33} is not a performance metric — it is a computational proof that eml and eml★ operate in categorically different function spaces on this target.

Reproducibility. Script: `eml_toolkit/direct_verification.py`. Requires NumPy only. Runtime < 1 second.

6. Summary

Result	Sec.	Status
Holomorphic closure of E	2	Unconditional theorem
Re, Im, conj ∉ E	2	Unconditional corollary
$\bar{z}$ = 1 − eml★(0, eml(z,1)) at depth 2	3	Conditional: Im(z) ∈ [−π, π)
Domain Im(z) ∈ [−π, π) (half-open)	3	SymPy + interval verification
|z| reachable at depth 4	3	Unconditional corollary
Density of {eml, eml★, 1} in C(K, ℂ)	4	Conditional on branch-safety lemma
Theorem 3.1 at machine precision (ratio 10^{33})	5	Numerical — fully reproducible

Open problems. (1) Does E ∩ ℝ coincide with the exp-log closure of {1} over ℚ? This is connected to the Schanuel conjecture. (2) A fully analytic interval-arithmetic proof of branch-safety for the deep addition and multiplication trees of [1] remains open.

Appendix A — Explicit Clean Arithmetic Blocks

The following constructions realize subtraction, negation, and 1/2 as finite eml-trees. Branch safety verified at 100 decimal digits on 10^5 random points.

subtract(x, y) — Depth 4:

```
eml( eml(1, eml(eml(1,x), 1)), eml(y, 1) )
```

minus(x) — Depth 7:

```
eml( eml(eml(1,eml(eml(1,x),1)), eml(eml(1,eml(eml(1,1),1)),1) ), 1 )
```

half — Depth 6 (constant):

```
eml(1, eml(1, eml(1, eml(1, eml(1, eml(1, 1))))))
```

Caveat. These blocks close branch safety for effective depth ≤ 4. Branch safety for the full addition ($K \approx 27$) and multiplication ($K \geq 17$) trees of [1] is verified numerically at 60 decimal places (zero violations, 1000 random points) but an analytic interval proof for arbitrary compacts remains open.

References

[1] A. Odrzywołek, All elementary functions from a single operator, arXiv:2603.21852v2 (April 2026).

[2] M. H. Stone, The generalized Weierstrass approximation theorem, Mathematics Magazine 21 (1948), 167–184.

[3] G. Terzo, Some consequences of Schanuel's conjecture in exponential rings, Comm. Algebra 36 (2008), 1171–1189.

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Ancillary files: branch_safety_final.py · verify_theorem4.py · eml_toolkit/direct_verification.py · eml_star_branch_safety_lemma_proof.pdf | License: MIT | Zenodo: 10.5281/zenodo.19183008 | GitHub: github.com/antparis/eml_star